Nasr Kasrin

Product Thinker | Architect | Hippie Software Engineer

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/nkasrin

A laid-back software architect/engineer that mixes passion and strategic vision. I shine on the intersection of engineering and product/people/cross-functional leadership *(see n42r.github.io)*

Positions

2021 – 2023 **Doctoral Candidate**, *University of Bamberg*, Germany
- Completed a PhD at the faculty of Information Systems & Applied Computer Science.
- Project: A data cataloging architectural pattern (data mesh) *(see n42r.github.io/phd)*

2015 – 2020 **Research Associate**, *SIMUTOOL Horizon2020 EU Project*, Germany
- Managed a project to develop a data lake / catalog for a manufacturing consortium.
- Architect, product manager, developer, team leader.
- Stack: Neo4j, RDF, Python, Flask, Web2py, Docker (multi-container), git, UML.

2011 – 2014 **R&D Engineer | Team Leader**, *TayalT*, Cairo, Egypt
- Developed AI players for games, resulting in improved app rankings (C++, Java).
- Managed multiple products (social media apps and games) and reported to the CEO.
- Coordinated cross-functional teams of technical, marketing, UI/UX, and strategy.

Skills

Databases	MongoDB, Neo4j, SQL, Google Firebase Cloud, RDF
Development	Python3 (pydantic, mypy, pylint), Docker (multi-container), Linux CLI, git, UML
Past Projects	Lisp, C++, PIC microcontroller programming (embedded)

Projects

2023 – 2024 **Muze**
AI-driven music discovery and recommendation service *(see github.com/n42r/virtmulib)*
- Uses LLMs (chatgpt/Llama), asks for user preferences and recommends music.
- **Results:** First pure-LLM service that does not use the Spotify recommendation engine.

2022 – 2023 **Guestrrday**
CLI tool for guessing year and label of songs *(see github.com/n42r/guestrrday)*
- **Results:** Successfully organized and tagged over 20,000 items.

2016 – 2019 **SIMUTOOL**, *EU Project*
Data catalog SaaS platform for an Industry 4.0 consortium *(see github.com/simutool)*
- Built interfaces, data formats, architecture; developer, supervisor, product manager.
- **Results:** Used to manage & exchange data assets between 8 companies for 3 years.

2012 – 2014 **Greetings Studio**, *TayaIT*
Social network for creating and sharing e-cards *(see n42r.github.io/gs)*
- Technical product management, product design, software architect.
- **Results:** Number of active users & rating steadily grew to 4 stars on the Apple Store.

2011 – 2013 **Tawla**, *TayaIT*
Backgammon board game App on the iOS mobile platform *(see n42r.github.io/tw)*
- Developed AI players and other components (ex., complex random number generators).
- **Results:** The rating of the App improved from 3 to 4+ stars citing the AI players.

2010 – 2011 **ArtSapiens**
R&D head of a team of 10+ to build software for the RoboCup 2D Soccer competition.
- **Results:** The software qualified and competed in the 2011 RoboCup finals.

2009 – 2010 **Cassie**, *Masters Project*
AI control software for agent/robot perception (Stack: LISP)
- **Results:** Discovered two novel methods, published 2 papers in top conferences.

2006 **Walk The Line**, *Competition (German University in Cairo)*
An Obstacle avoidance, maze-navigating vehicle (Stack: Micro-C PIC programming).
- **Results:** Selected for display in the annual open science day of the university.

2005 **Reversi3D**, *Competition (German University in Cairo)*
3D board game of Othello (Stack: Java v5 & Java3D).
- **Results:** One of the top three in a competition involving 100+ submitted games.

Education

2023 **PhD. (Dr. rer. nat.)**, *Information Systems & Applied Computer Science, Otto-Friedrich-Universität Bamberg*, Bamberg, Germany

2010 **MSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

2008 **BSc.**, *Computer Science & Engineering, The German University in Cairo*, Egypt

Publications

2021 Data-Sharing Markets for Integrating IoT Data Processing Functionalities. *CCF Transactions on Pervasive Computing and Interaction.*

2018 Semantic Data Management for Experimental Manufacturing Technologies. *Datenbank-Spektrum.*

2010 Focused Belief Revision as a Model of Fallible Relevance-Sensitive Perception. *Künstliche Intelligenz 2010: Advances in Artificial Intelligence.*

2010 High-Level Perception as Focused Belief Revision. *European Conference on AI (ECAI).*

Languages

Arabic	Mother Tongue
English	Fluent
German	Good understanding, Basic Speaking (B1, Telc Certificate)